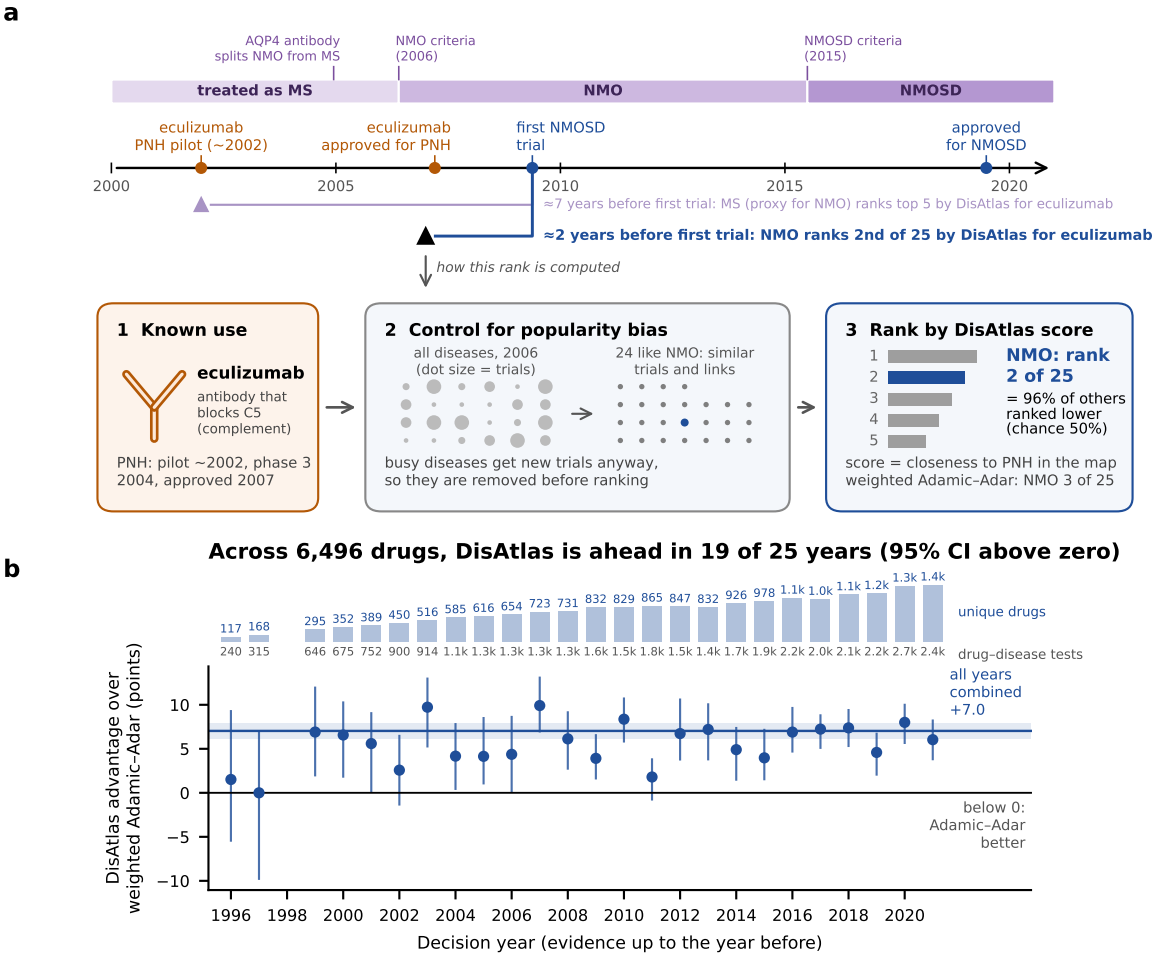

a
AQP4 antibody splits NMO from MS
NMO criteria (2006)
NMOSD criteria (2015)
treated as MS
NMO
NMOSD
eculizumab PNH pilot (~2002)
eculizumab approved for PNH
first NMOSD trial
approved for NMOSD
2000
2005
2010
2015
2020
≈7 years before first trial: MS (proxy for NMO) ranks top 5 by DisAtlas for eculizumab
≈2 years before first trial: NMO ranks 2nd of 25 by DisAtlas for eculizumab
how this rank is computed
1 Known use
eculizumab
antibody that blocks C5 (complement)
PNH: pilot ~2002, phase 3 2004, approved 2007
2 Control for popularity bias
all diseases, 2006 (dot size = trials)
24 like NMO: similar trials and links
busy diseases get new trials anyway, so they are removed before ranking
3 Rank by DisAtlas score
1
2
3
4
5
NMO: rank 2 of 25
= 96% of others ranked lower (chance 50%)
score = closeness to PNH in the map
weighted Adamic–Adar: NMO 3 of 25
b
Across 6,496 drugs, DisAtlas is ahead in 19 of 25 years (95% CI above zero)
117 168
240 315
295 352 389 450 516 585 616 654 723 731 832 829 865 847 832 926 978 1.1k 1.0k 1.1k 1.2k 1.3k 1.4k
646 675 752 900 914 1.1k 1.3k 1.3k 1.3k 1.3k 1.6k 1.5k 1.8k 1.5k 1.4k 1.7k 1.9k 2.2k 2.0k 2.1k 2.2k 2.7k 2.4k
unique drugs
drug-disease tests
all years combined +7.0
below 0: Adamic–Adar better
DisAtlas advantage over weighted Adamic–Adar (points)
10
5
0
−5
−10
1996 1998 2000 2002 2004 2006 2008 2010 2012 2014 2016 2018 2020
Decision year (evidence up to the year before)